

Introduction to Python

Module 1 — Getting Started

Variables, data types, and your first code

Your Very First Python

The traditional first program just prints a message. In Python it is one line:

```
print("Hello, world!")
```

`print()` is a **function**: it displays whatever you put between its parentheses. The text in quotes is a **string**. Run it, and Python shows:

```
Hello, world!
```

That is the loop you'll repeat forever: write code, run it, read the output.

Variables: Naming Things

A **variable** is a name that holds a value. You create one with = (“assign the value on the right to the name on the left”).

```
name = "Denise"  
age = 35  
  
print(name)  # Denise  
print(age)   # 35
```

You can reuse and update variables. The name lets you refer to the value later without retyping it.

Naming rules: use letters, numbers, and underscores; start with a letter; no spaces. Choose **descriptive** names — `word_count`, not `wc`.

The Basic Data Types

Every value has a **type**. The four you'll use constantly:

Type	What it is	Example
int	whole number	42, -7, 0
float	decimal number	3.14, -0.5, 2.0
str	text (a "string")	"hello", 'Python'
bool	true / false	True, False

You can check any value's type with the `type()` function:

```
type(42) # <class 'int'>
```

Numbers & Arithmetic

Python is a capable calculator. The operators:

```
7 + 3 # 10 (add)
7 - 3 # 4 (subtract)
7 * 3 # 21 (multiply)
7 / 3 # 2.33 (divide)
```

```
7 // 3 # 2 (floor div)
7 % 3 # 1 (remainder)
7 ** 3 # 343 (power)
```

Note / always gives a float. // and % (“modulo”) are handy more often than you’d expect — e.g. testing if a number is even with `n % 2 == 0`.

Working with Strings

Strings are text, in single or double quotes. You can **combine** them and insert variables:

```
first = "computational"
second = "text analysis"

# join with +
first + " " + second  # "computational text analysis"

# f-strings: insert variables with {}
f"I study {first} {second}"
```

f-strings (note the `f` before the quote) are the modern way to build text with variables inside — you'll use them constantly.

Converting Between Types

Sometimes you need to change a value's type — e.g. a number typed as text into an actual number. Use `int()`, `float()`, `str()`:

```
int("42")    # 42 (str -> int)
str(42)      # "42" (int -> str)
float(3)     # 3.0 (int -> float)
```

Watch out: `"5" + 3` is an **error** — you can't add text to a number. You must convert first: `int("5") + 3`. Mixing types is one of the most common beginner bugs.

Module 1: Takeaways

- ✓ `print()` displays output; that's your window into what code does
- ✓ **Variables** (`name = value`) hold values under a descriptive name
- ✓ Four core **types**: `int`, `float`, `str`, `bool`
- ✓ Python does **arithmetic** with `+` `-` `*` `/` `//` `%` `**`
- ✓ Build text with `+` and **f-strings**; **convert** types with `int()` `float()` `str()`

Now open the notebook to try all of this yourself — then Module 2 covers **data structures** (lists and dictionaries).